PPIA HOMEWORK 1: Anonymizing the Google Local Reviews Dataset

Group members:
Ivan Radovanovic, 2223575 - Laszlo Kosa, 2183964 - Maxime Henry, 2210555

The setting

We use the Google Local Reviews dataset from the McAuley Lab at UC San Diego, collected from public Google Maps pages up to September 2021 and split into one file per US state. We use only Vermont: the reviews, plus the category and price level of each business, linked through the business ID. Vermont has a good balance: most places only have a few reviews, so reviewers are easier to single out, but there is still enough data to show the risks and the effect of anonymization. We also added one made-up attribute, the amount spent during the visit, to have a financially sensitive value to protect.

The data

For Vermont, the dataset comes in two files: a review file and a business (metadata) file. Both are in JSON lines format, which means that every line of the file is one record.

The review file contains 852,203 reviews. Some places were downloaded twice, so 10,225 reviews appear two times. After removing these duplicates, we have 841,978 reviews, written by 288,212 different users about 11,243 businesses. Every review has the following attributes:

- user_id: the number of the reviewer's Google account
- name: the reviewer's name as shown on Google
- time: the exact time when the review was posted
- rating: the number of stars, from 1 to 5
- text: the text of the review
- pics: links to the photos added to the review
- resp: the owner's reply, if there is one, with its own text and time
- gmap_id: the ID of the business, which connects the review with the business file

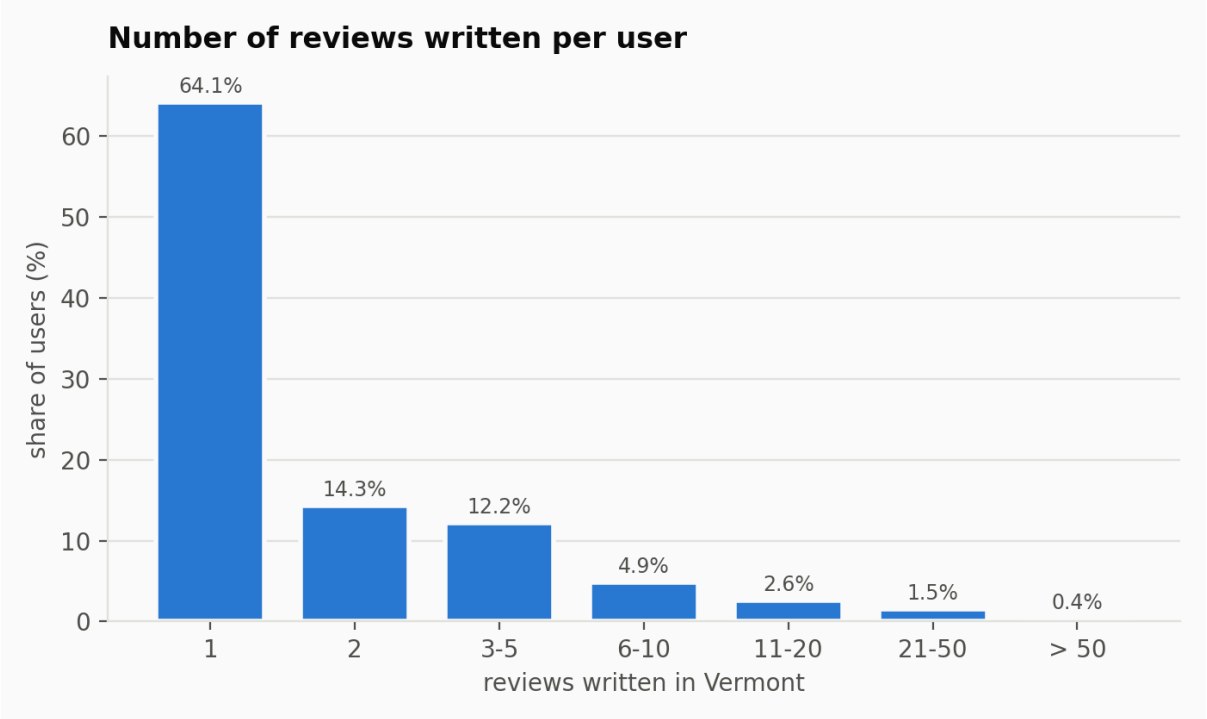
The business file has one record for every place, with its name, address, coordinates, description, list of categories, average rating, number of reviews, price level ($ to $$$$), opening hours, other information (like accessibility or pet-friendly), and a list of related places. From this file, we only use the categories and the price level. The rest of the business information is public on Google Maps anyway and is not needed for our analysis.

Besides the original attributes, we created one synthetic attribute, "spent": the amount of money the reviewer spent during the visit. The value is random, but it depends on the type of place, so it stays realistic (for example, a coffee is cheaper than a hotel night).

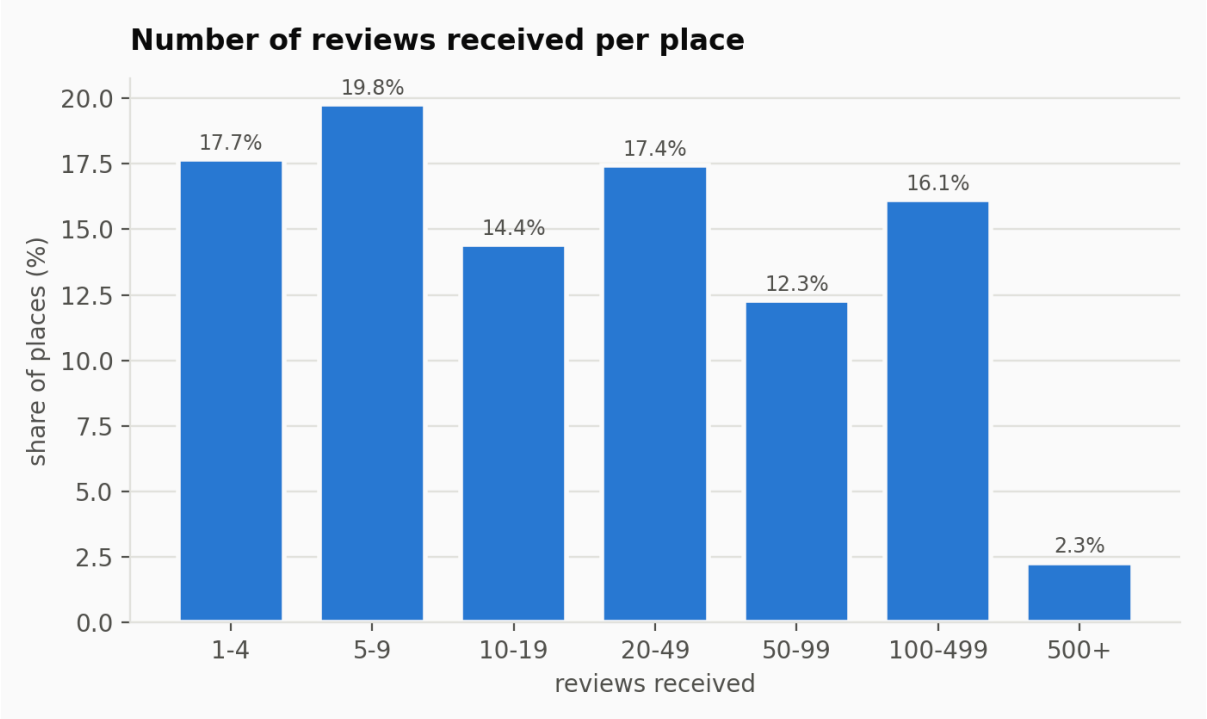
We chose this data because it combines things that are typical for real personal data: a direct link to real people (account and name), information about where and when they went, free text that they wrote themselves, and, with the spent attribute, a financial value. At the same time, it is large enough to give meaningful statistics.

Below are descriptions of the main distributions.

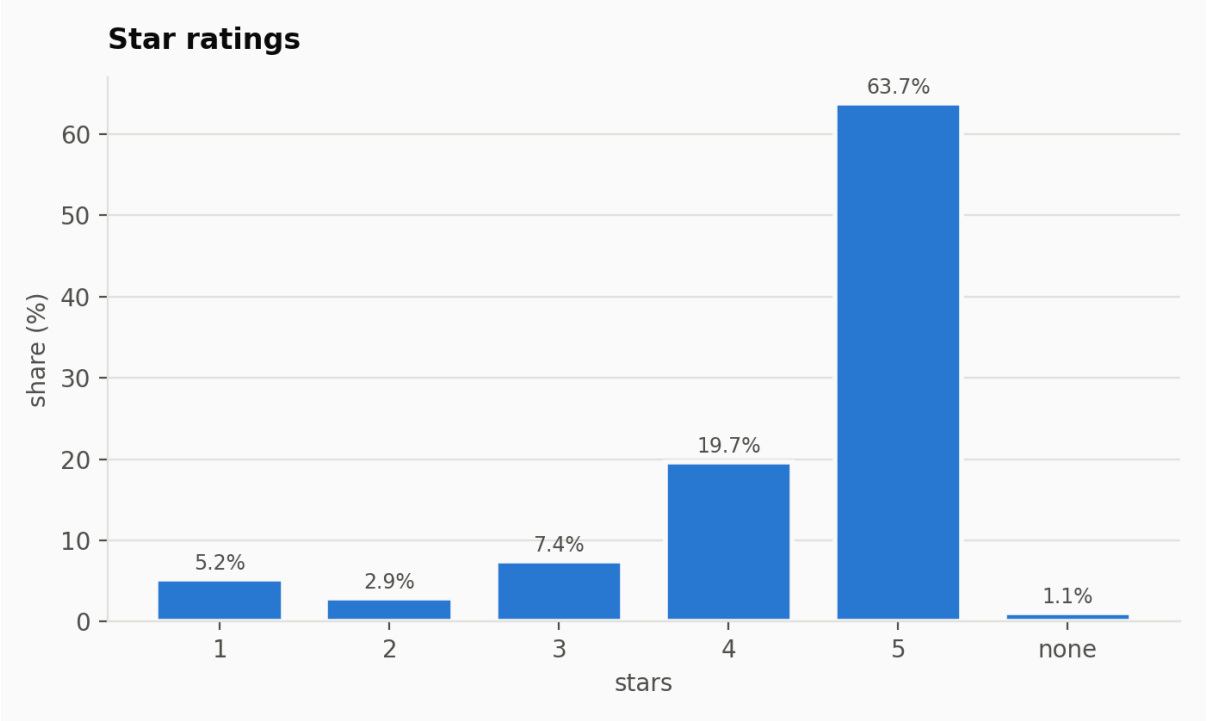
Users and places. Most people write very few reviews: about two-thirds of the users (64%) wrote only one review in Vermont, while a small group of very active users wrote dozens or hundreds (and one account more than 9,000).



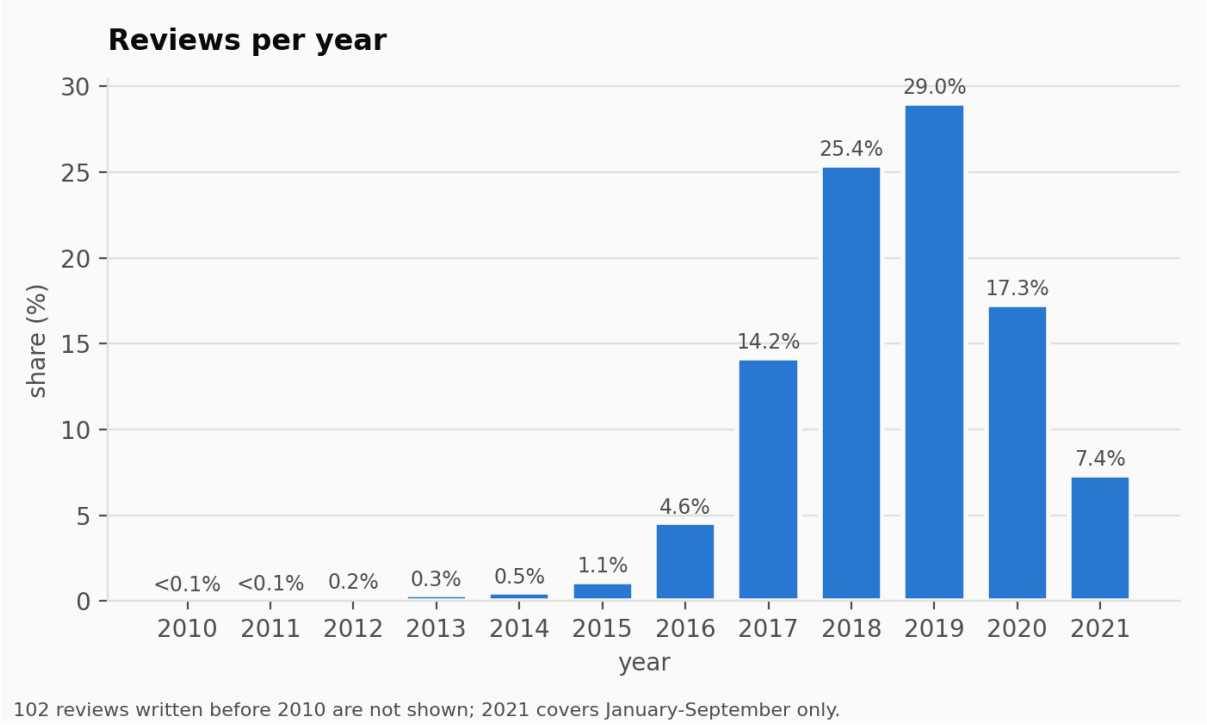
Places are also very uneven: a typical place has around 18 reviews and more than a third of them have fewer than 10, while around 60 popular places have more than a thousand.



Rating. Ratings are mostly positive: almost two-thirds of the reviews (63.7%) give 5 stars, and about 20% give 4 stars. Only about 1% of the reviews have no rating.

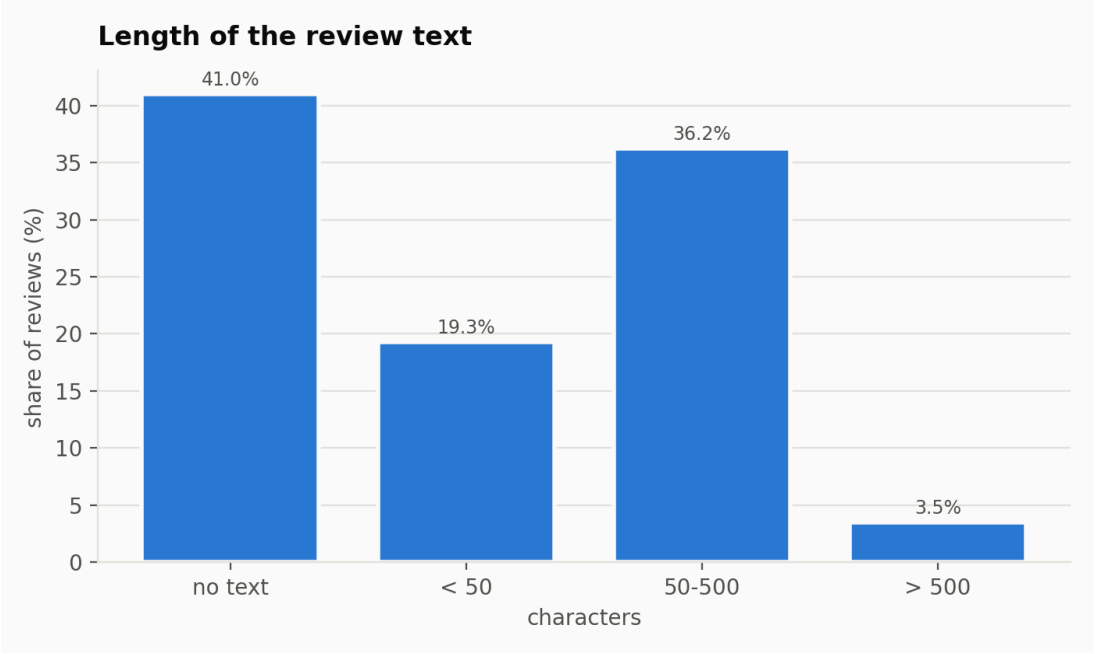


Time. The reviews go from 1999 to September 2021, but almost all of them are recent: most were written between 2017 and 2020, with the peak in 2019.

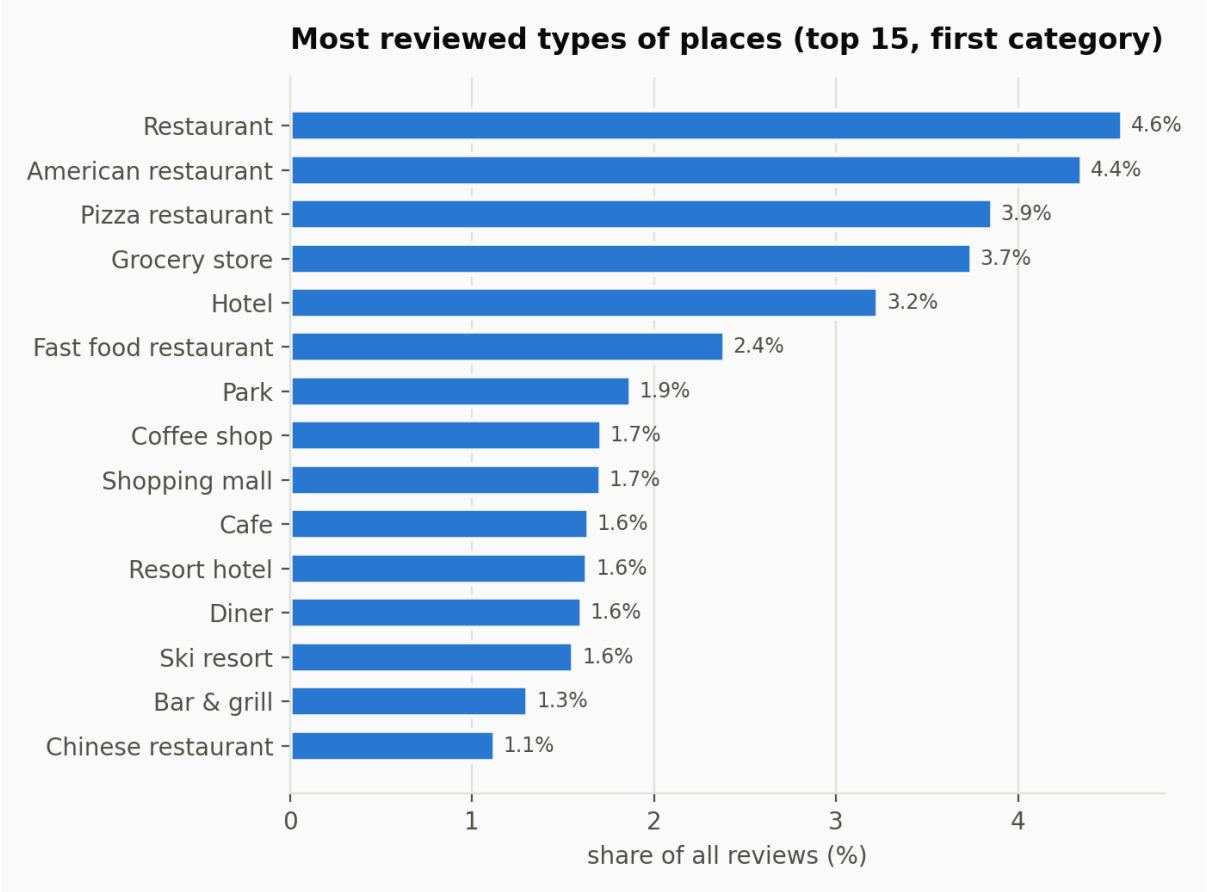


Text, photos, and replies. About 41% of the reviews have only a rating and no text. Most of the reviews that do have text are short to medium length, and only 3.5% of all reviews are longer than 500

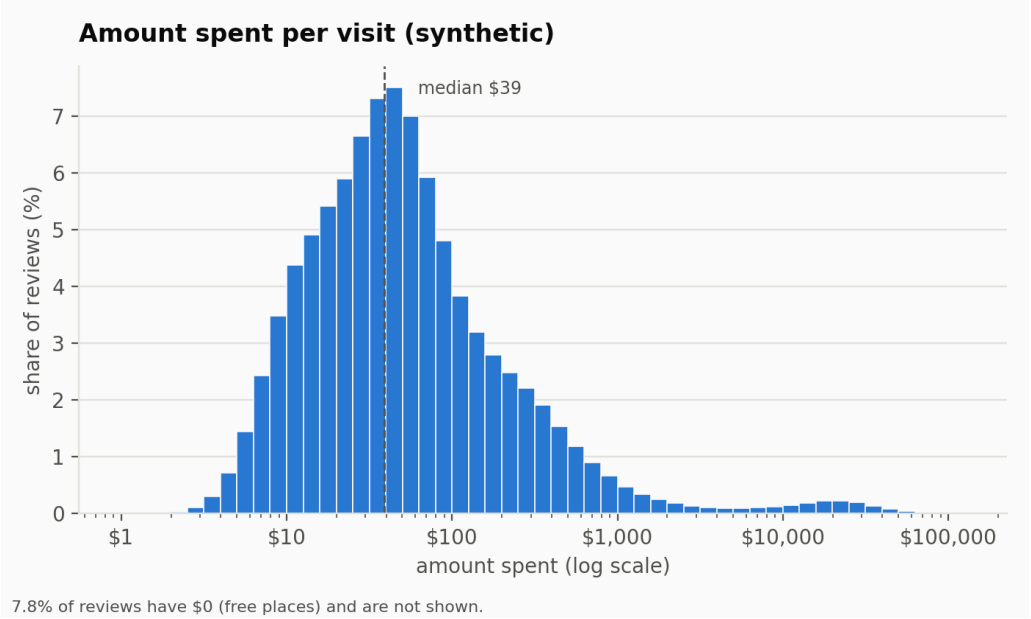
characters. Photos are rare (3.5% of reviews), and the owner replied to about 11.5% of the reviews.



Categories and price level. Places have about two categories each on average, out of around 2,000 different categories. The most reviewed types of places are restaurants, grocery stores, and hotels. Some categories are sensitive in themselves, for example, places related to health (doctors, dentists, clinics, pharmacies, about 2% of the reviews) or religion (churches). The price level is only known for less than half of the reviews and is almost always $ or $$.



Amount spent. Since spending depends on the type of place, its values are very spread out: about 60% of the visits cost between $10 and $100 (the median is about $39), around 8% cost nothing (parks, churches, etc.), and a few are very large, such as car purchases above $10,000.



Risks and threats

Once the setting has been described, provide an intuition of what could be the risks arising in the setting, which data are exposed, and which information might put the user in danger. Importantly: 1) do not limit yourself only to the actual data/attributes available in the dataset, but consider operating in the real scenario where these data were collected originally; 2) consider also the risks that might arise from external record linkage.

Imagine publishing the review dataset for anyone to download, for example to support research on recommender systems, and we assume that the business metadata (name, address, coordinates, category, price level, opening hours) is fully public. The goal is that, once the data are public, the people who wrote the reviews remain private. Reviews are already visible on Google Maps, but there they are scattered across pages, tied to a platform and removable by their authors. On the contrary, a release turns them into a complete, structured table that can be searched and manipulated. In terms of the EDPB guidelines, the raw data allows a person to be singled out, their records to be linked, and sensitive facts to be inferred.

Name and user_id (direct identifiers)

The name is the reviewer's real name in most cases, and the user_id is a unique number that is linked to the contributor's profile. It also joins all the reviews of one person, exposing the individual to record linkage.

Timestamp (quasi-identifier)

The exact moment of a review is very precise, there are no two identical timestamps in the dataset. In most cases, the date of the review is quite close to the actual visit. The main danger, however, is linkage: any other source with comparable time information (another scrape, a geotagged social media post, a booking) can be matched with it. The time of the owner's reply adds little about the reviewer but helps to match the two sides of an exchange.

Place (gmap_id) and the public metadata
Since the metadata is public, we have no control over it, and the gmap_id is the key that connects the two datasets. The category could expose confidential attributes if it is not handled with care. For example, health and religion might get exposed, causing attribute disclosure. The price level and the type of places visited also suggest spending power and lifestyle. Because the metadata stays public, the data of individuals can be kept safe by "protecting" the businesses.

Rating
This is the attribute that has the highest value over danger ratio, so it is advised to keep as much of it as possible. This does not necessarily mean that it has no privacy concerns, here, uniqueness causes a very serious possibility of a privacy leak.

Review text
This is the richest and least controlled attribute, because nothing in the schema tells what it contains. People might write about their family and children, with ages, about their health, work, or other sensitive information. In some cases, they name staff members or owners, thus exposing a third party and some of their personal details. This field causes too many unique use cases, thus, it is practically impossible to effectively sanitize it.

Photos
Photos can show faces (including those of children and bystanders), documents, or receipts. Even without metadata, the content itself can be matched by face or image search with a social profile.

Owner's reply
Replies very often start with the reviewer's name and restate details of the visit or service. This confirms the identity of the reviewer and their relationship with the business. Replies also name employees and may contain the owner's contact details. Just like in the case of the review text, the same scenario is faced here: it is practically impossible to effectively sanitize it.

After the release
In the original scenario, Google Maps, the author is a real account, often with a real name and photo. Reviewers wrote for the customers of a business, not for researchers or profilers, so a release goes beyond the context in which they agreed to share. A published copy does not coincide with the right to erasure, so it needs to be properly sanitized. Attackers also get stronger over time: The bigger an individual's digital footprint is (and with time it certainly grows), the easier it is to recognize their patterns of activity.

External record linkage
Outside of the connection to the metadata, an attacker could:

- follow the user_id to the public profile and connect it to activities and reviews in other states,
- match name, town and month with social networks, professional profiles or public registers and gather sensible personal information,
- join the public addresses of places such as hotels and home services with rental or property records.

To sum it up, without proper precautions, an attacker could gain access to basically any kind of information about the victim without limit.

Utility

Describe what the utility of your data is (e.g., can it be used to train an SVM for classification? Is it important to use it to carry out regression? etc.). Once the utility has been defined, describe the utility when using the original (sensitive) data to carry out the task (e.g., what is the AUC if you build a simple classifier using the clear data? What about the R2 in case of a linear regression?).

Our dataset is meant for research on reviews and local businesses, for example, recommender systems or the study of customer satisfaction. As a utility measure, we chose a simple question: is the

length of a review related to its rating, in other words, do unhappy customers write more? We use the original data, in which we have the length of every review.
The answer is a clear yes. The average rating stays high for short reviews and drops quickly once a review becomes long.

Review length (characters)	Average rating	Negative (1–2 stars)
no text	4.39	5.2%
1–49	4.48	5.3%
50–100	4.46	6.4%
101–200	4.36	9.9%
201–500	4.13	16.9%
501–1,000	3.60	31.5%
over 1,000	2.93	48.9%

Reviews with no text or fewer than 100 characters average about 4.4–4.5 stars and only 5–6% of them are negative, while reviews longer than 1,000 characters average 2.93 stars and almost half of them are negative.
This is one of the utilities we want to keep after anonymization: we publish the length classes of the reviews (short, medium, and long), thus keeping the intended correlation.

Implementing privacy-preserving solutions

Show the effects of implementing the anonymization techniques that we have seen during the lecture. In particular, show both the positive and negative effects in terms of privacy protection, e.g., use some of the privacy measures that we have discussed to quantify the risks, discuss changes in utility, e.g., show how the performance compares with respect to the original data, and their tradeoff, e.g., through ru-maps.
Feel free to report snippets of code, figures, and everything necessary to illustrate your case.
[Max 9000 characters]
Anonymization part :
We started by classifying the attributes of our dataset as we saw in the lecture. The user_id and the name are identifiers, they allow us to univocally identify the person, so they have to go. The place, the date, and the amount spent are quasi-identifiers, they don't say much alone, but combined they can identify a person. And the review text and the owner's response are confidential attributes, as they can contain the opinions and sometimes the names of real people.

We decided to remove the user_id and name completely from the dataset as these are personal identifiers. These are primary identifiers and could severely compromise the privacy of the individuals who originally posted the reviews. In the lecture, this is called sanitizing, removing the explicit identifiers. Also, to reduce the risk of crossing time data with other public datasets, we removed the hour and day when the Google review was posted. So in the end we kept the month and year to have an idea of the age of the comment, and cutting the day out of the date is a generalization, one step up the hierarchy. To avoid complete data loss, we computed the difference of time between the review and the owner's response when there was one. The idea was to have an idea of the owner's social media activity and to allow research into whether more engagement on social media from the owner correlated with better reviews and more satisfactory experiences for visitors.

```
# anonymize_v2.py
row.pop('time') # the hour is an explicit time stamp, drop it
row['date'] = row['date'][:7] # generalization: YYYY-MM-DD -> YYYY-MM
```

We replaced the reviews with their length at first, as people usually mention family members by name in the reviews. We could have used an LLM-as-a-judge, for example, to rate the “positiveness” or not of the comment, but that was out of scope for this homework. We did the same for the owner's response, as there was also sensible data, where owners usually responded to reviews by addressing the user by name. After that, review lengths were put into 3 categories to avoid identifying reviews by their length with other public datasets. These categories are: under 50 characters, between 50 and 500, and over 500.

```
# anonymize_v3.py
def length_category(n):
 if n is None: return None
 if n < 50: return 'short' # carries little information
 if n <= 500: return 'medium' # the informative core
 return 'long' # verbose outliers
```

We believe that the most useful comments are between these two limits. Indeed, reviews under 50 characters don’t provide much useful information to a user, and those over 500 will surely not be read by the user.

The last attribute we needed to protect was the amount spent. This amount is not in the raw data, we generated it ourselves with a model based on the type of place, because people don't spend the same at a restaurant and at a car dealership.
To anonymize it, we first applied the 5% rule. Inside each type of place, every amount below the 5% quantile was raised to it, and every amount above the 95% quantile was capped at it.

```
# anonymize_v4.py - bottom and top coding at the 5%/95% quantiles
def clip_to_quantiles(spent, q05, q95):
 return min(max(spent, q05), q95)
```

The idea was to remove the extreme values that identify a review on their own, for example, a single review with a 22,000$ amount is rare enough to be unique. After that, we rounded the amounts to a list of values going from 0 to 10,000$.

```
# anonymize_v4.py - rounding on a ladder of published points
TIERS = [0, 5, 10, 25, 50, 100, 250, 500, 1000, 2000, 5000, 10000]
def round_to_tier(spent):
 return min(TIERS, key=lambda t: abs(spent - t))
```

Then the values holding less than 5% of the reviews were merged with the closest one next to it, so that no published value is rare enough to identify someone. In the end, we have only seven values left : 0, 10, 25, 50, 100, 250, and 500 dollars.

The most “dangerous” quasi-identifier left was the place itself. Indeed, many Vermont businesses only have a few reviews, so knowing the place is often enough to find the person behind the review. So we applied k-anonymity with k = 5, every combination of quasi-identifiers we publish must be shared by at least 5 reviews, otherwise the group is small enough to match a review to a single person. We did it by generalization, we replaced the Google id of the place with its categories. This way, two places with the same categories are not distinguishable anymore. We used a ladder in three steps to be sure that every group of places we publish is shared by at least k = 5 reviews, and the last step is what the lecture calls a local suppression, we replace the place information with a missing value, but we keep the review itself.

```
# kanonymity.py - generalize first, suppress the cell as a last resort
if tuple_rows[category_key(cats)] >= k: # full category list
```

```
 groups[g] = (sorted(cats), 0)
elif first_rows[cats[0]] >= k: # first category only
 groups[g] = ([cats[0]], 1)
else: # local suppression
 groups[g] = (None, 2)
```

If the full list of categories is already shared by 5 reviews or more, we keep it, this is the case for 99.85% of the reviews. Otherwise, we only keep the first category of the place, and for the last rare cases we remove the place information completely (1,031 reviews, including the 192 whose place has no category). No review was deleted, and the smallest group we publish has exactly 5 reviews. In the end, we also removed the state attribute because we are exclusively working on the state of Vermont.

Discussion, conclusion and open challenges

Detail any relevant additional information, open challenges, risks, and possible follow-ups. [max 1500 characters]

We built a step-by-step anonymization pipeline that turns the raw reviews into a dataset ready to be published, while keeping all the reviews. We classified the attributes as it was presented in the lecture, removed the direct identifiers, generalized the date to the month, and replaced the review and reply texts with their length classes, thus keeping the content hidden. The owner's reply delay was calculated and kept, however.

For the amount spent, bottom and top coding at the 5% and 95% quantiles of each type of place, rounding, and the merging of rare values were done. In the case of the place, which is the most telling quasi-identifier, we applied k-anonymity with k = 5 through generalization, meaning that the place's gmap_Id was replaced by its categories. On a few occasions, local suppression was needed.

In conclusion, the result supports the analyses of ratings, time trends, spending, and owner presence across types of places, while making the link back to an individual reviewer much harder. As follow-ups, the created sanitization and anonymization pipeline could be extended with ℓ-diversity and t-closeness.

GenAI disclosure

Detail any use of GenAI

Data sanitization (v1). After our first team meeting, we agreed on a list of mandatory operations: remove the user ID and name; convert the timestamp into a readable date and time; keep the rating; replace the text, photos, and owner's reply by their length/count, a has-response flag, and the response delay; look up each place's categories in the metadata file through the gmap_id; add a synthetic amount spent; and add the state. We gave this list and the dataset to the AI, which wrote and ran the Python script anonymize_v1.py. Some details were decided by the AI and checked by us: exact duplicate reviews (10,225) are removed, times are converted from UTC to Vermont local time, and response delays that come out negative (reviews edited after the reply) are set to null. The later steps reduced the date to month and year (v2).

Synthetic spending. The rules for the spent attribute were designed by the AI: each type of place gets a typical amount (e.g., ~$14 at a coffee shop, ~$250 at a hotel, $0 at a park), adjusted by the place's price level when known. The actual values are drawn by Python's random module, not generated by the AI itself.

Other uses. The AI also wrote the script that produces the figures in "The data" section, computed descriptive statistics, and reviewed our anonymization scripts for possible issues. All results, numbers, and text were checked and edited by the team.

Group members contribution

Ivan Radovanovic:

- Found the dataset
- Worked on initial sanitization of the dataset
- Wrote The setting and The data sections

Name1 Surname1: list of contributions by Name1 Surname1
Name2 Surname2: list of contributions by Name2 Surname2